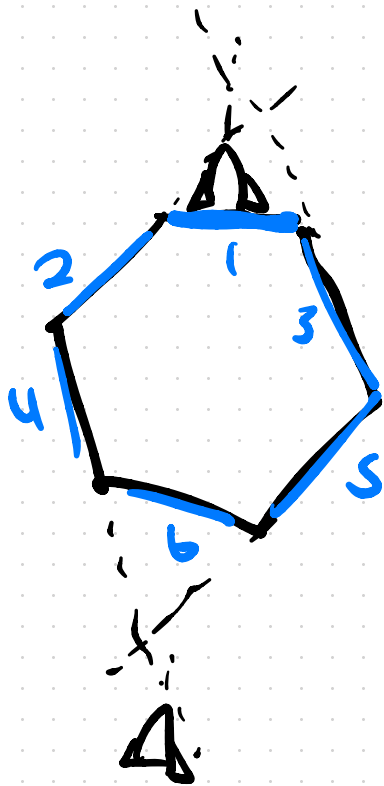
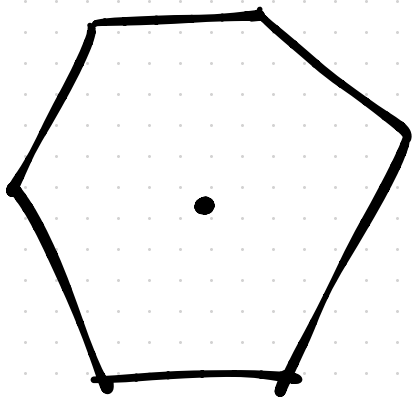


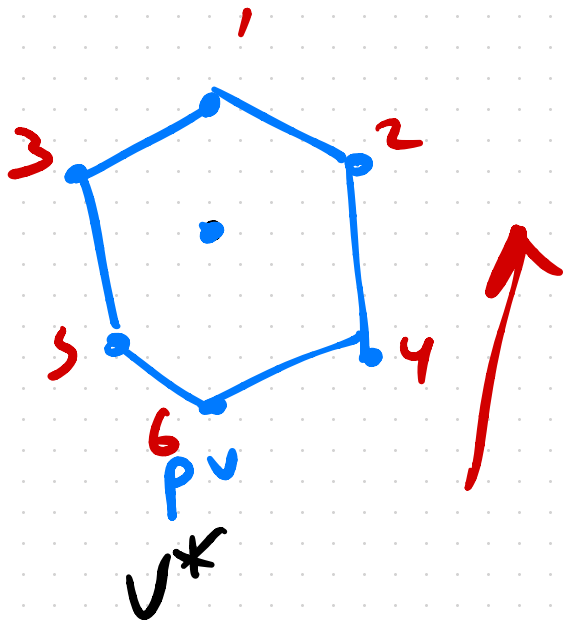
Thm: If Δ is the boundary of a simplicial polytope, then it has a shelling.





P

V



If we have a shelling $F_1, \dots, F_n$ of Δ , and S_k is the set of faces that attach to k $(d-1)$ -faces during the shelling, then $|S_k| = h_k$ where $(h_0, h_1, \dots, h_{d-1})$ is the h -vector of Δ .

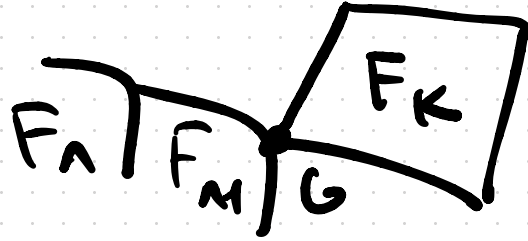
Cor: If Δ is shellable, then $h_k \geq 0$ for all k .

Prop: If $F_1, \dots, F_n$ is a shelling of a simplicial sphere, then $F_n, \dots, F_1$ is also a shelling.

Proof: Suppose $F_n, \dots, F_1$ is not a shelling. Then for some k ,

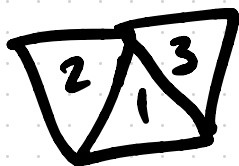
$$F_k \cap \left(\bigcup_{i=k+1}^n F_i \right) = \emptyset$$

is not pure of dim $d-1$.



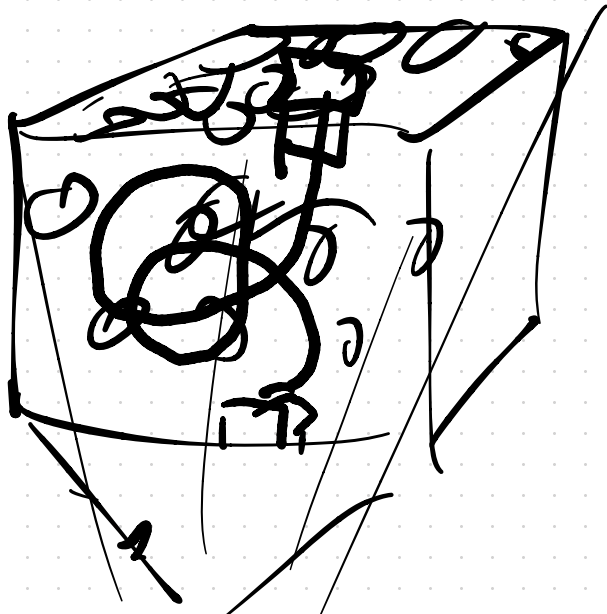
The complement of $F_1 \cup \dots \cup F_k$ is $F_1 \cup \dots \cup F_{k-1}$.

However, $F_1, \dots, F_{k-1}$ is not a shelling order at the link of G .

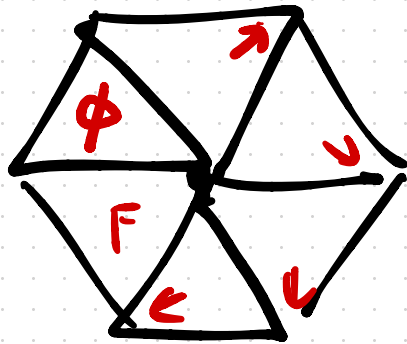


Thm (Rudin): There is a non-shellable triangulation of the tetrahedron in $\mathbb{R}^3$.

Thm: There is a non-shellable topological triangulation of the 3-sphere.



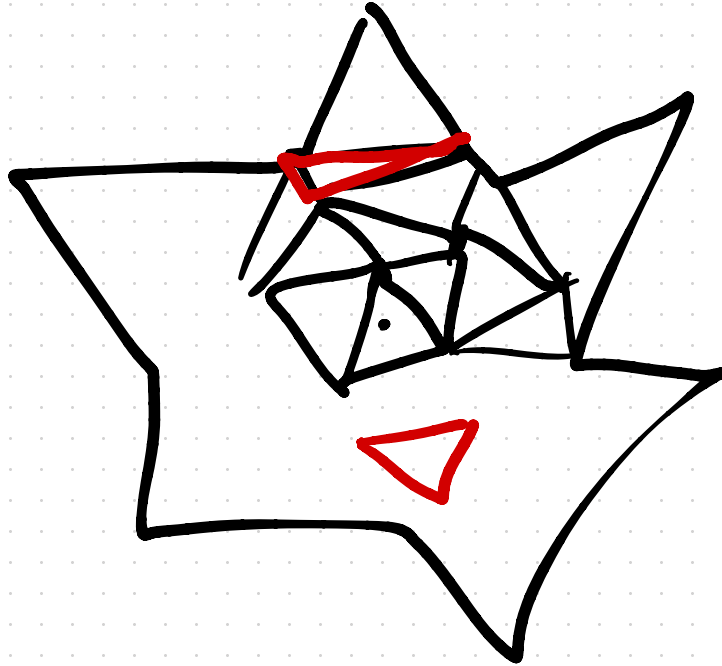
A ^{pure} simplicial complex Δ is **partitionable** if for each facet F there exists a face $\sigma_F \subset F$ such that $\{\sigma_F, F\}$ is a partition of Δ .



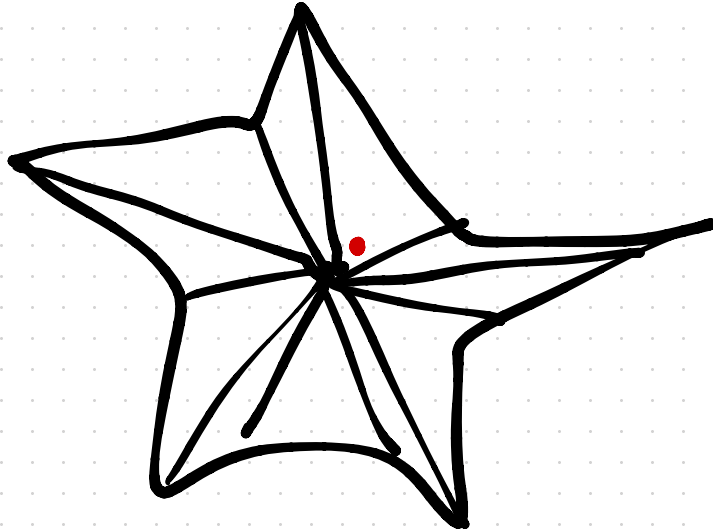
Observation: If Δ is shellable, then it is partitionable.

Prop: Every triangulation of a star-shaped ball is partitionable.

Proof:



Cor : The boundary of a star-shaped ball is partitionable.



Upper bound
theorem

What is the maximum number of k -faces that a d -polytope with n vertices can have?

$$f_k \leq \binom{n}{k+1}$$

This bound is tight for $k \leq \lfloor d/2 \rfloor$.

$$h_k(\Delta) = \sum_{i=-1}^{k-1} (-1)^{k-1-i} \binom{d-i}{k-1-i} f_i(\Delta)$$

$$h_{\frac{d-1}{2}} = h_{\frac{d+1}{2}}$$

$$h_{\frac{d-3}{2}} = h_{\frac{d+3}{2}}$$

The f -vector is determined by $f_0, f_1, \dots, f_{\lfloor d/2 \rfloor - 1}$.
 However, the coefficients are not nonnegative.

Thm (McMullen): For all k , the maximum number of k -faces of a d -polytope with n vertices is given by any simplicial $\lfloor d/2 \rfloor$ -neighborly polytope with n vertices.

Note:

$$f_i = \sum_{k=0}^{d+1} \binom{d+1-k}{i+1-k} h_k$$

So it suffices to show neighborly polytopes simultaneously maximize all the h_k .